Thus we expect that the ⟦BAOAB⟧ scheme recovers $\mathcal{O}(h^4)$ configurational averages in the limit of infinite friction. This scheme requires no extra force evaluations, and this particular feature appears to simply be a serendipitous result of the form of the correction function.

This is a property common to no other Langevin splitting scheme that we are aware of. Though curiously its conjugate method, the ⟦OABAO⟧ scheme, has the property

$$\int_{\mathrm{Q}} f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega_q = 0,$$

implying that we obtain high-order averages for functions purely of the momenta in the limit of infinite friction. However, as the momenta are always Gaussian distributed in canonical sampling, we are far more interested in functions of position to high accuracy (and sampling $\exp(-\beta U(\boldsymbol{q}))$ rather than the Gaussian-distributed momenta).

It is important to note that this superconvergence property is only present in the limit of large time, i.e. this is an attribute of the invariant distribution of the discretization schemes $\hat{\rho}$. As such, there are no bonuses to the weak or strong order of the superconvergent schemes, though equally there is no added cost to using the methods.

All of the symmetric splitting schemes discussed are very easy to implement relative to other popular Langevin integrators, as many codes already implement the standard velocity Verlet ⟦BAB⟧ style integration. An exact Ornstein-Uhlenbeck solve can be added where it is required in the scheme in order to effectively integrate Langevin dynamics: then setting $\gamma = 0$ reduces the integrator to a standard Verlet scheme, integrating deterministic Newtonian dynamics.

7.10 The Limit Method

We now consider schemes in the limit $\gamma \to \infty$, where the exact solution of the vector OU process reduces to redrawing momenta from the canonical distribution, so $\boldsymbol{p}_{n+1} = \sqrt{k_\mathrm{B}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}$, where $\mathbf{R}$ is a vector of i.i.d. normal random numbers. Alternatively, we could consider the limit of the particle mass going to 0, although this requires a reformulation of Langevin dynamics (7.4) so that the friction is proportional to the velocity instead of the momentum [233]. Whichever limit is taken, we would expect the ultimate result to be the same. (Here we have reintroduced the masses in order to present the method, since they may be useful scaling parameters in simulation.)

The first order schemes of the form ⟦XYZ⟧, as well as schemes such as ⟦XOYOX⟧ are not consistent in this overdamped limit: the errors in computed